Gigabyte miniature factory

A byting look at the making of some Giga boards!

During the Taipei Computex 2010, we got a chance to check out one of Gigabyte's motherboard manufacturing plants. We bring you all the dope on the process.

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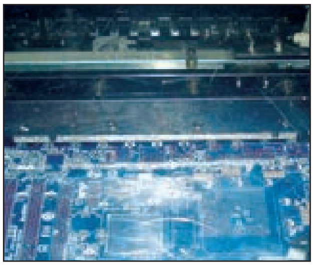
Gigabyte makes great motherboards – a fact nobody can deny. So when we were given a chance at a peep inside their Nan-Ping factory, that assembles and manufactures motherboards, graphic cards, server components and embedded solutions, we naturally jumped at the opportunity. The Nan-Ping plant is an eight-storey building, with 45,000 square meters of floor space and is located in rural Taiwan, about an hour away from Taipei by road.

Inside the plant, we were greeted in a huge foyer, and given footwear to wear on top of our shoes. There is a beautiful miniature model of the factory made totally of motherboard components, including expansion slots and heatsinks.

Before entering we passed through an air shower to remove dust particles. There was a brief conference explaining the process. Gigabyte doesn't manufacture PCBs. They receive the PCBs in a semi-finished state. After this, manufacturing is a four-stage process.

Conductive solder gets on to the PCB here

SMT process: Acronym for Surface Mount Technology, this stage, as you might guess, is where the conductive layer of solder paste is printed on the bare PCB. After this tiny components like resistors and voltage regulators are mounted. Core components like main chipsets (Northbridge, Southbridge) are mounted next. The machine doing this is called a "component gun", and can mount components very, very fast. This is actually one of the main stages of the production, where all the finer aspects (read smaller components) are taken and mounted.



After conductive soldering

Obviously, each machine needs to be calibrated and set up for a particular model of motherboard. As motherboards are made in batches, at one time, you will only see one particular motherboard model being made. After a batch of identical motherboards are churned out, the machines need to be recalibrated, which is an automated job done by software, to handle another model of motherboard.

Shown above is what the motherboard looks like after conductive soldering is done.

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